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Waterproof Microphone Device

Abstract

The present invention provides a waterproof microphone device installed on a vehicle body. The vehicle body includes a frame having a circular opening. The waterproof microphone device includes an outer housing, a circuit board, a microphone, a cover, a first waterproof membrane, and a casing. An inner wall of the frame forming the opening gradually slopes inwardly from an outer surface to an inner surface of the frame. The casing includes a first wall, a second wall, and an outer sidewall. The outer sidewall gradually slopes inwardly from the first wall to the second wall to form an inclined annular sound channel between the inner wall of the frame enclosing the opening and the outer sidewall of the casing. In present invention, the waterproof microphone device can achieve better waterproof effect, and has a more stable structure with a beautiful appearance.

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Background/Summary

FIELD OF THE PRESENT INVENTION

[0001] The present invention relates to the field of microphone, and more particularly, to a waterproof microphone device in vehicle.

DESCRIPTION OF RELATED ART

[0002] With the development of technology, the application of microphones in vehicles is becoming increasingly widespread. In related art, the surface of the vehicle will be equipped with through holes as sound holes, and the microphone will be installed inside the vehicle. A waterproof membrane will be applied on the microphone, and the microphone will be connected to the outside of the vehicle through the sound holes. However, this structure will cause the release of turbulent energy and generate strong wind noise during the driving state, which will affect the reception of the microphone.

[0003] When washing the vehicle or in rainy weather, when the water flow vertically impacts on the surface of the vehicle, the gas inside the sound hole is violently compressed, which can easily cause damage to the internal structure of the microphone. The waterproof membrane will also be damaged by directly impacted of water flow. In addition, water stagnation near the sound hole can easily cause blockage, and freezing in cold weather is more likely to cause external pickup function failure. Moreover, when washing a vehicle with hot water, water vapor will penetrate the waterproof membrane and condense in the internal channels, making it difficult to discharge.

[0004] Therefore, it is desired to provide a new waterproof microphone device which can overcome the above problems.

SUMMARY

[0005] In view of the above, the embodiments of the present invention provide a new waterproof microphone device which has a stable structure and good waterproof effect.

[0006] The present invention provides a waterproof microphone device. The waterproof microphone device is installed on a vehicle body. The vehicle body includes a frame having a circular opening. The waterproof microphone device includes an outer housing, a circuit board disposed in the outer housing, a microphone mounted on the circuit board, a cover connecting the outer housing and the frame, a first waterproof membrane sealing on the cover, and a casing fixed with the cover. An inner wall of the frame forming the opening gradually slopes inwardly from an outer surface to an inner surface of the frame. The casing includes a first wall facing out of the vehicle body, a second wall opposite to the first wall, and an outer sidewall connecting the first wall and the second wall. The outer sidewall gradually slopes inwardly from the first wall to the second wall to form an inclined annular sound channel between the inner wall of the frame enclosing the opening and the outer sidewall of the casing.

[0007] As an improvement, the casing includes at least one recess portion extending from the second wall to the outer sidewall of the casing.

[0008] As an improvement, wherein a number of recess portions is multiple, and adjacent recess portions spaced equally apart from each other on the outer sidewall.

[0009] As an improvement, wherein the cover includes a main body portion covered with the outer housing and a protruding portion arranged on the main body portion, the protruding portion provided with a receiving groove communicated to the sound channel, and the main body portion provided with a first through-hole communicated to the receiving groove.

[0010] As an improvement, wherein the receiving groove includes a central portion and at least three extension portions diffused outwardly from the central portion, the first through-hole

arranged relative to one of the extension portions.

[0011] As an improvement, wherein the receiving groove has a T-shaped structure, a Y-shaped structure, a cross shaped structure, or a flower shaped structure.

[0012] As an improvement, wherein a surrounding wall of the protruding portion forming the receiving groove is an arc-shaped structure.

[0013] As an improvement, wherein the waterproof microphone device further includes a sealing ring located between the cover and the circuit board and a second waterproof membrane arranged between the sealing ring and the main body portion of the cover, the sealing ring provided with a second through-hole communicated to the first through-hole of the cover.

[0014] As an improvement, wherein the circuit board is provided with a third through-hole communicated to the second through-hole, the microphone located on a lower surface of the circuit board.

[0015] As an improvement, the cover is provided with a plurality of installation columns extending from the protruding portion, the installation columns penetrating through the first waterproof membrane and fixedly connected to the casing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Many aspects of the exemplary embodiments can be better understood with reference to the following drawing. The components in the drawing are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present invention. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

[0017] FIG. 1 is an illustrative isometric view of a waterproof microphone device in accordance with one embodiment of the present invention.

[0018] FIG. 2 is a partly exploded view of the waterproof microphone device of FIG. 1.

[0019] FIG. 3 is an exploded view of the waterproof microphone device of FIG. 1.

[0020] FIG. 4 is an exploded view of the waterproof microphone device from another perspective of FIG. 1.

[0021] FIG. 5 is an illustrative cross-sectional view of the waterproof microphone device taken along line A-A of FIG. 1.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0022] The present invention will hereinafter be described in detail with reference to exemplary embodiments. To make the technical problems to be solved, technical solutions and beneficial effects of the present invention more apparent, the present invention is described in further detail together with the figures and the embodiments. It should be understood the specific embodiments described hereby is only to explain the disclosure, not intended to limit the disclosure.

[0023] Referring to the FIGS. 1-5, the present invention provides one embodiment of a waterproof microphone device **100**. The waterproof microphone device **100** is installed on a vehicle body **200**. The vehicle body **200** includes a frame **202** having a circular opening **201**. The waterproof microphone device **100** includes an outer housing **1**, a circuit board **2** disposed in the outer housing **1**, a microphone **3** mounted on the circuit board **2**, a cover **4** connecting the outer housing **1** and the frame **202**, a first waterproof membrane **5** sealing on the cover **4**, a casing **6** fixed with the cover **4**, a sealing ring **7** located between the cover **4** and the circuit board **2**, a second waterproof membrane **8** arranged between the sealing ring **7** and cover **4**, and a plurality of connector pins **9**.

[0024] The out housing **1**, the circuit board **2**, the microphone **3**, the cover **4**, the first waterproof membrane **5**, the sealing ring **7**, the second waterproof membrane **8**, and the connector pins **9** are disposed in the vehicle body **200**. The waterproof microphone device **100** can be installed on the

vehicle body **200** near the car door, car lights, or other locations.

[0025] An inner wall **2021** of the frame **202** forming the opening **201** gradually slopes inwardly from an outer surface **2022** to an inner surface **2023** of the frame **202**. That is, an aperture of the opening **201** gradually decreases from the outer surface **2022** to the inner surface **2023**. As the opening **201** is a circular or an elliptical shape, the inner wall **2021** is a smooth curved inclined plane.

[0026] The casing **6** is a circular or an elliptical shape corresponding to the opening **201**. The casing **6** includes a first wall **61** facing out of the vehicle body **200**, a second wall **62** opposite to the first wall **61**, and an outer sidewall **63** connecting the first wall **61** and the second wall **62**. The first wall **61** of the casing **6** is flush with the outer surface **2022** of the frame **202**. The outer sidewall **63** gradually slopes inwardly from the first wall **61** to the second wall **62**, forming an inclined annular sound channel **60** between the inner wall **2021** of the frame **202** enclosing the opening **201** and the outer sidewall **63** of the casing **6**. In this way, when the vehicle is in motion, an airflow flows through the annular sound channel **60**, resulting in less turbulent energy and lower wind noise.

[0027] The casing **6** includes at least one recess portion **64** extending from the second wall **62** to the outer sidewall **63** of the casing **6**, and the recess portion **64** does not extend to the first wall **61** of the casing **6**. A number of recess portions **64** is multiple, and adjacent recess portions **64** are spaced equally apart from each other on the outer sidewall **63**. The surface that forms the recess portion **64** is also a smooth curved surface, and the recess portions **64** ensure a sufficient width of the sound channel **60**, at the same time, the recess portion **64** reduces high-frequency resonance.

[0028] The out housing **1** includes a first containment portion **11** for accommodating the circuit board **2**, the microphone **3**, the sealing ring **7**, and the second waterproof membrane **8**, and a second containment portion **12** separated from the first containment portion **11**. The out housing **1** further includes a stop wall **13** separating the first containment portion **11** and the second containment portion **12**. One end of each connector pin **9** is electrically connected to the circuit board **2**, and the other end of the connector pin **9** extends into the second containment portion **12**.

[0029] The circuit board **2** is supported on an inner wall of the out housing **1**. The microphone **3** is located on a lower surface of the circuit board **2**. The microphone **3** is a MEMS microphone or other types of microphones.

[0030] The cover **4** and the first waterproof membrane **5** are covered on the out housing **1** and sealed the first containment portion **11**. The cover **4** includes a main body portion **41** covered with the outer housing **1** and a protruding portion **42** arranged on the main body portion **41**. The main body portion **41** has a substantially rectangular shape, and the protruding portion **42** has a substantially circular shape. The protruding portion **42** is provided with a receiving groove **421** communicated to the sound channel **60**, and the main body portion **41** is provided with a first through-hole **411** communicated to the receiving groove **421**.

[0031] The receiving groove **421** includes a central portion **4211** and at least three extension portions **4212** diffused outwardly from the central portion **4211**. The first through-hole **411** is arranged relative to one of the extension portions **4212**. The receiving groove **421** has a T-shaped structure, a Y-shaped structure, a cross shaped structure, or a flower shaped structure, etc. That is, the shape formed by the central portion **4211** and the extension portion **4212** is T-shaped, Y-shaped, cross shaped, or flower shaped, etc. The surrounding wall of the protruding portion **42** forming the receiving groove **421** is an arc-shaped structure.

[0032] When washing the vehicle or on rainy days, water flows enter from one extension portion **4212** of the receiving groove **421**, and the generated air pressure is released from another extension portion **4212** of the receiving groove, avoiding the damage to the microphone **3**. At the same time, the water flows pass through the annular sound channel **60**, reducing the impact on the first waterproof membrane **5**. Due to the instability of water flows in the annular hole, the water flow is easy to detach under the push of airflow or gravity, therefore, the residual liquid in the receiving

groove **421** can also flow out in any direction of the sound channel **60**, thereby avoiding the retention of the liquid. During vehicle operation, the continuous flow of gas in the sound channel **60** promotes the evaporation and discharge of condensed water, while also avoiding the retention of the liquid.

[0033] The cover **4** is provided with a plurality of installation columns **43** extending from the protruding portion **42**, and the installation columns **43** penetrates through the first waterproof membrane **5** and fixedly connected to the casing **6**, thereby achieving the installation of the casing **6**. An installation groove **412** is provided on one side of the main body **41** of the cover **4** away from the protruding portion **42**, and the installation groove **412** is communicated with the first through-hole **411**. The sealing ring **7** is installed inside the installation groove **412**.

[0034] The sealing ring **7** is provided with a second through-hole **71** communicated to the first through-hole **411** of the cover **4**. The second waterproof membrane **8** locates between the second through-hole **71** of the sealing ring **7** and the first through-hole **411** of the cover **4**.

[0035] The circuit board **2** is provided with a third through-hole **21** communicated to the second through-hole **71**. The process of collecting sound is that, the sound entering from the sound channel **60** and sequentially passing through the receiving groove **421**, the first through-hole **411**, the second through-hole **71**, and the third through-hole **21**, and finally reaching the microphone **3**.

[0036] Comparing with the related art, the present invention provides a waterproof microphone device. The waterproof microphone device is installed on a vehicle body. The vehicle body includes a frame having a circular opening. The waterproof microphone device includes an outer housing, a circuit board disposed in the outer housing, a microphone mounted on the circuit board, a cover connecting the outer housing and the frame, a first waterproof membrane sealing on the cover, and a casing fixed with the cover. An inner wall of the frame forming the opening gradually slopes inwardly from an outer surface to an inner surface of the frame. The casing includes a first wall facing out of the vehicle body, a second wall opposite to the first wall, and an outer sidewall connecting the first wall and the second wall. The outer sidewall gradually slopes inwardly from the first wall to the second wall to form an inclined annular sound channel between the inner wall of the frame enclosing the opening and the outer sidewall of the casing.

[0037] In present invention, when the vehicle is in motion, an airflow flows through the annular sound channel, resulting in less turbulent energy and lower wind noise. When washing the vehicle or on rainy days, water flows enter from one extension portion of the receiving groove, and the generated air pressure is released from another extension portion of the receiving groove, avoiding the damage to the microphone. At the same time, the water flows pass through the annular sound channel, reducing the impact on the first waterproof membrane and extended its service life. The residual liquid in the receiving groove can also flow out in any direction of the sound channel, thereby avoiding the retention of the liquid. During vehicle operation, the continuous flow of gas in the sound channel promotes the evaporation and discharge of condensed water, while also avoiding the retention of the liquid. At the same time, the waterproof microphone device of this invention has a beautiful appearance.

[0038] It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiment, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

Claims

1. A waterproof microphone device, the waterproof microphone device installed on a vehicle body, and the vehicle body comprising a frame having a circular opening, the waterproof microphone

device comprising: an outer housing; a circuit board disposed in the outer housing; a microphone mounted on the circuit board; a cover connecting the outer housing and the frame; a first waterproof membrane sealing on the cover; and a casing fixed with the cover, the casing comprising: a first wall facing out of the vehicle body; a second wall opposite to the first wall; and an outer sidewall connecting the first wall and the second wall; wherein an inner wall of the frame forming the opening gradually slopes inwardly from an outer surface to an inner surface of the frame, and the outer sidewall gradually slopes inwardly from the first wall to the second wall to form an inclined annular sound channel between the inner wall of the frame enclosing the opening and the outer sidewall of the casing.

2. The waterproof microphone device as described in claim 1, wherein the casing comprises at least one recess portion extending from the second wall to the outer sidewall of the casing.

3. The waterproof microphone device as described in claim 2, wherein a number of recess portions is multiple, and adjacent recess portions spaced equally apart from each other on the outer sidewall.

4. The waterproof microphone device as described in claim 1, wherein the cover comprises a main body portion covered with the outer housing and a protruding portion arranged on the main body portion, the protruding portion provided with a receiving groove communicated to the sound channel, and the main body portion provided with a first through-hole communicated to the receiving groove.

5. The waterproof microphone device as described in claim 4, wherein the receiving groove comprises a central portion and at least three extension portions diffused outwardly from the central portion, the first through-hole arranged relative to one of the extension portions.

6. The waterproof microphone device as described in claim 5, wherein the receiving groove has a T-shaped structure, a Y-shaped structure, a cross shaped structure, or a flower shaped structure.

7. The waterproof microphone device as described in claim 5, wherein a surrounding wall of the protruding portion forming the receiving groove is an arc-shaped structure.

8. The waterproof microphone device as described in claim 4, wherein the waterproof microphone device further comprises a sealing ring located between the cover and the circuit board and a second waterproof membrane arranged between the sealing ring and the main body portion of the cover, the sealing ring provided with a second through-hole communicated to the first through-hole of the cover.

9. The waterproof microphone device as described in claim 8, wherein the circuit board is provided with a third through-hole communicated to the second through-hole, the microphone located on a lower surface of the circuit board.

10. The waterproof microphone device as described in claim 4, wherein the cover is provided with a plurality of installation columns extending from the protruding portion, the installation columns penetrating through the first waterproof membrane and fixedly connected to the casing.
